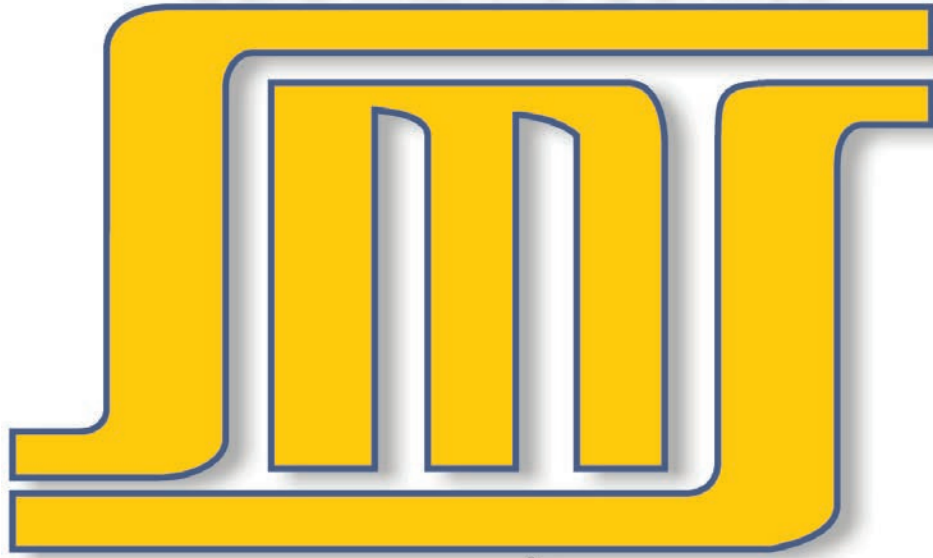


Sure Marine Service®



www.suremarine.com

TSL-17

Installation Guide

With SureWire™ Control





TSL-17 Installation and Information

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June 8th 2017

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Visit our showroom to see and feel these heaters in operation.



The text, drawings and other artwork are not to scale and are simply intended to be guides to aid in your selection and installation process. They do not replace manufacturer or owners manuals. We will provide operation restrictions for the unit, but certain areas have legal restrictions that must be heeded as well.

Ensure all parts of the heater's installation are within approved standards and workmanship.

Non-approved components may not meet operational requirements of this heating system and may result in failure of that component or of the entire system.

The content of this document is furnished for informational use only. It is subject to change without notice. Sure Marine Service assumes no responsibility or liability for any errors or inaccuracies that may appear in the content.

The information in this packet should be read and followed. Any mechanical problem due to failure to follow these installation requirements will not be covered under warranty.

This manual in no way removes you from any legal obligation that may be present in your area, nor does it make SMS liable for any installation done by our customers. Reference sources that should also be consulted (though much of this manual has done this already) are ABYC standards, NFPA codes and USCG standards.



This symbol indicates a hazardous situation which, if not avoided, will result in death or serious injury.



This symbol indicates information that, if not carefully followed, can cause minor injury or damage to equipment!

SAFETY

BEFORE ATTEMPTING ANY PROCEDURE IN THIS BOOK, READ AND UNDERSTAND ALL THE SAFETY INFORMATION CONTAINED THEREIN. IN ADDITION, ENSURE ALL INDIVIDUALS WORKING WITH YOU ARE ALSO FAMILIAR WITH THESE SAFETY PRECAUTIONS.

When installation of the unit is completed but before operation, the entire system should be inspected. SMS will provide this inspection or recommend a dealer to do one.

TSL-17 Installation and Information



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TSL-17 Installation and Information

Properly Positioning Heating System Components

Exhaust Routing

Water System

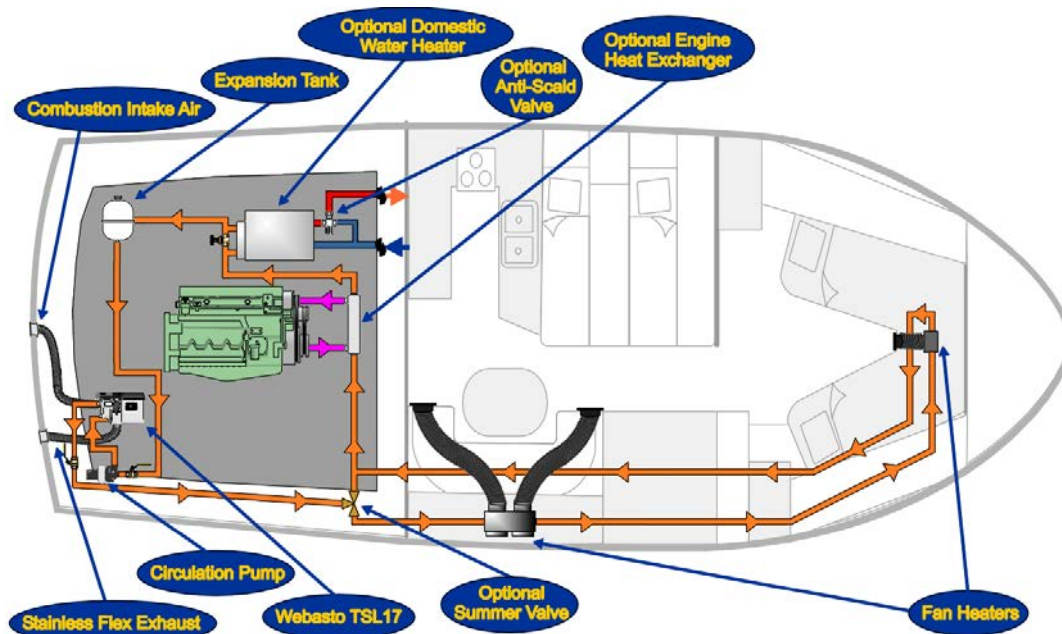
It is important for the proper operation of your heating system that the expansion tank is located above all components in your water system. This is to maintain head pressure on the system and to prevent air from interfering with the flow of water in the system. If any component is located above the expansion tank, air may be trapped in this component and restrict the flow of water through the system. Reduced flow can lead to poor performance of the heating system and potential damage to the water pump.

Your layout should have an unrestricted flow of water from the expansion tank to the system pump. Components that may restrict the flow of water such as fan heaters or hot water tanks should not be placed between the expansion tank and the pump. Components that do not restrict the flow of water such as engine heat exchangers may be placed in this line.

A direct line should also exist between the pump and the heater. The pump should have no components except free-flow shut-off valves between itself and the boiler.

Fan heaters can be installed anywhere in the system, provided they are installed below the expansion tank and not in the line from the expansion tank to the pump or boiler.

Like the fan heaters, heat exchangers must be below the expansion tank. However, because they are free-flowing they may be installed in between the expansion tank, system pump and heater.



Marine Layout Example

Expansion Tank



The expansion tank must be located at the highest point of the heating system and connected directly to the inlet of the circulation pump.



This kind of system must allow for expansion and contraction of system components as they heat up and cool down. The expansion tank accommodates this by allowing the water level to rise and fall while still maintaining head pressure on the system pump. At the same time, the expansion tank acts as a "burp" tank, allowing for air in the system to escape.

System Circulation Pump

For proper operation 90 degree fittings must not be mounted on or near either suction or discharge sides of the pump.

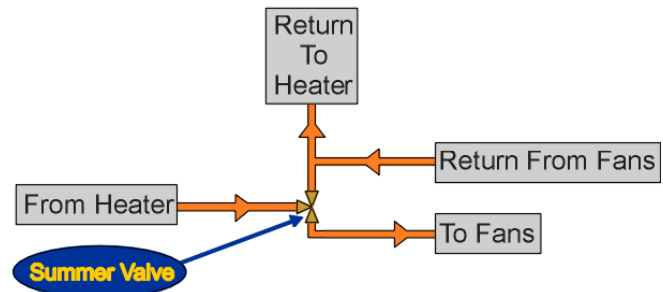
The system pump must be free from air for good circulation and proper water flow through the system. Air bubbles in the pump can cause cavitation that will impede the flow of water out of the pump. Many factors can lead to pump cavitation: improper installation, air trapped in components such as fan heaters, or improper seals in the water system.



Maximum recommended single loop hose run is 100'.

Summer Valve

During the summer you may want to use a "Summer Valve" to avoid circulating hot water through your heating system while making your domestic hot water. A bypass (Summer) valve will allow water to pass through your domestic water heater and not through the cabin heating loops. This allows you to make hot water without heating the boat.



Domestic Water Heaters

A domestic water heater should be on a loop that will always be on. If the domestic water heater is installed in a zone that is shut off for any reason, it will no longer have the ability to be heated by the boiler.

An exception to the rule that nothing may be installed between the expansion tank, pump and heater is the Everhot brand domestic water heater. This water heater provides minimal restriction to the flow of the coolant through it.

A diesel boiler can provide VERY HOT water.



Whenever using a diesel boiler to heat domestic water it is necessary to install an Anti-Scald valve (also known as a thermostatic mixing valve).

Failure to do so may cause serious injury.

Fan Heaters

Fan heaters should be spaced throughout the boat based on the BTU demands of each area to be heated.

All living spaces should be heated to ensure an even, efficient heat for your boat. An unheated room will draw heat from the system unless it is always sealed from heated areas. This can cause drafts and result in uneven heating.



REAL 6022

To figure the BTU demand of a space, measure the volume (Length x Width x Height) taking in to consideration the contour of the boat. Once volume is figured, multiply the cubic feet by 12 if the space is below deck, or by 15 if the space is above deck or has many windows. This figure should give you an idea of the BTU demand for this space.

Once the BTU demands are figured, fan heater models can be chosen. In most cases, space for fan heater installation guides the choice as much as the heat demand. A single unit may meet the demands for a space, or several units may be required. Further, when one unit may suffice, there may not be enough room for its installation and multiple smaller fan heaters may be needed. When choosing where to put the fan heaters, ensure that the heat will be ducted into the space evenly. If a space requires multiple fan heaters, try not to install every fan heater in one location, arrange them to evenly to heat the space.

As you select the fan heater locations be aware of where the hose must run, and in which zone you want the fan heater. Also, be sure that the location will meet the manufacturer's installation requirements. To avoid spreading dangerous or noxious fumes all fans should draw either from the cabin or outside air.

Thermostat(s)

Mercury thermostats are not to be used. They are easily affected by motion and vibration. They will cause short cycling and intermittent running of the heater. If you decide to use a thermostat other than the one we have provided, contact SMS to find a compatible model.

These thermostats are low voltage, low current devices. They can signal a relay to power the fan heaters.

When deciding where to mount the thermostat, we advise temporarily wiring it on an extended cable, and testing it in various locations to find the best location for proper heating of your boat. This will help you find the spot that is least effected by things like drafts and direct sun which can fool the thermostat.



REAL 843x



SC1600B

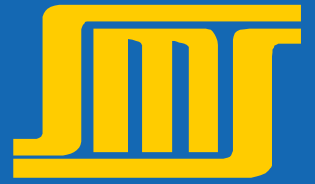


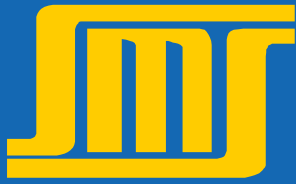
REAL 6400



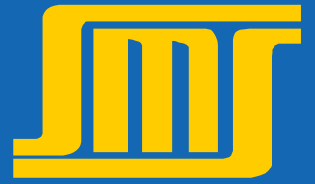
Exhaust System

Exhaust System





Exhaust System



Fuel System

The boiler uses a single line (supply only) fuel system.

Restrictive components may affect the operation of the unit. All fittings in the system should be accessible for maintenance and inspection and should be sealed with fuel-rated pipe thread sealant to prevent leaks.

Teflon tape should never be used to seal a fuel system, it will void your warranty. Most tape is not rated for fuel and small pieces may enter the system and foul the burner.



Per ABYC standards, a shut-off valve must be installed at the tank.

Isolation valves should also be installed for future maintenance of the heater or fuel filter. These valves should be free-flow valves that do not restrict the fuel flow.



Do not tee into an already existing fuel line that is installed for something else such as an engine line or generator line.

The fuel pump cannot compete with any other system for fuel. We advise either installing a dedicated line to the fuel tank, or installing a separate day tank for the unit. It may be possible to tee into a fuel manifold that does not have any suction in the line. SMS has fuel siphon tubes available for direct installation into the tank.

The Heater can be supplied from an auxiliary diesel fuel tank. This way a gas powered vessel can take advantage of diesel heating.



SMS stocks 2, 3, and 5 gallon aluminum fuel tanks.



Fuel Pump Information:

The pump has arrows stamped on it to indicate direction of fuel flow.

Neither the fuel pump nor the heater may be installed more than 6 ft. (2 meters) below the top of the fuel tank.

Maximum length of the fuel system should not exceed 30 feet (10 meters).

Maximum lift from the bottom of the dip tube to the pump is 3 ft. (1 vertical meter).

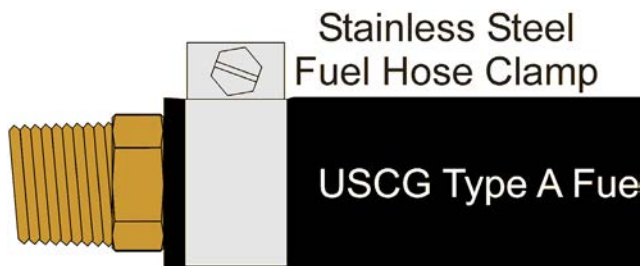
Maximum rise after the pump is 9 ft. (3 vertical meters).

It should be within 4 feet of tube from the tank

Exceeding these specifications may result in erratic operation of the heater or pump.

Installing The Fuel Lines

As you install the fuel line mount all components permanently assuring they are liquid and vapor tight. All hose connections must use corrosion resistant Stainless Steel clamps over the hose barb.



Per ABYC standards, all fuel lines should be supported by clamps every 12 inches attached to the bulkhead with corrosion-resistant fasteners and of a design to eliminate cutting or abrading action where in contact with the fuel line. SMS offers nylon clamps designed specifically for fuel hose or metal tube.



Each component of the system must be independently mounted. The fuel system cannot support the weight of components such as filters or valves.



Fuel Filter

Due to unpredictable quality of fuel and the possibility of foreign matter entering the burner, a fuel filter must be installed in the fuel supply.

SMS recommends the model WDL60046 Delevan filter.



Like any other component in a fuel system, the fuel filter must be mounted so it is accessible for changing and maintenance and it must be independently installed on a bulkhead.



Webasto has issued an advisory stating that bio-diesel has been causing premature wear resulting in non-warranty annual replacement of the burner. There is no warranty for damage caused by poor quality fuel or non approved mixtures of bio-diesel.

If you ignore this warning and choose to run bio-diesel, the fuel pump will also need more attention. Running the heater a minimum of 20 minutes every 2 weeks will help prevent the fuel pumps solenoid from sticking.

Be sure to **use only 3/16 copper or stainless steel tubing** for the fuel line for the heater. Larger diameters do not work for the dosing pump. Rubber hose can be used when joining copper tube and an appliance together. Copper fuel line must not contact any part of a metal hull. In a metal hulled boat either use sleeved copper or stainless steel fuel line.

When installing metal fuel line, be sure to check for knurling in the tube from cutting. This should be reamed out with a tubing cutter. Do not allow shavings to fall into the tubing.

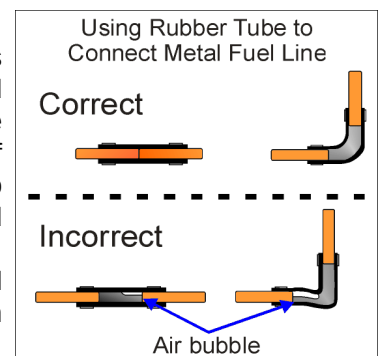
Fuel Pump Installation

Refer to the diagrams provided for fuel pump installation restrictions. Use the mounting kit provided with the pump for vibration isolation.

Install the 3/16" copper tubing from the tank, to the filter and then to the fuel pump. This run should have no loops or kinks that may affect the operation of the fuel system. The tube should be secured with mounts to stable bulkheads or frames. You should install isolation valves for future fuel filter replacement and for pump maintenance.

Included with the heater kit is a fuel filter that should be installed between the tank and fuel pump. Larger fuel filters can be installed to filter more contamination and extend the time between filter changes. These filters are available from Sure Marine.

The heater kit includes sections of fuel hose and hose clamps. These can be used to join sections of copper fuel line and to connect the fuel pump and heater to the copper tubing. Do not over tighten the fuel hose clamps, you may crush the fuel line or nick the hose.



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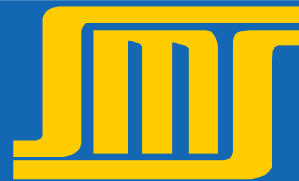


Fuel System





Electrical System





Thermostat(s)

Room thermostats must have snap action contact points to prevent point bounce during rolling conditions. The milli-volt type of thermostat MUST NOT be used.

Mercury thermostats are not to be used, they are easily affected by motion and vibration. They will cause short cycling and intermittent running of the heater. If you decide to use a thermostat other than the one we have provided, contact SMS to find a compatible model.

These thermostats are low power signaling devices. The SureWire™ board controls the boiler and can manage 3 amps per heating zone. Use relays to control higher current or higher voltage devices.

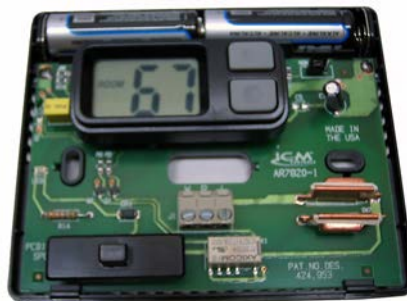
When deciding where to mount the thermostat, we advise temporarily wiring it on an extended cable, and testing it in various locations to find the best location for proper heating of your boat. This will help you find the spot that is least effected by things like drafts and direct sun which can fool the thermostat.



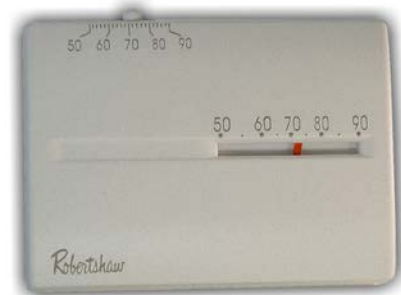
SC1600L-White



Aube TH135



SC1600B



W001-467

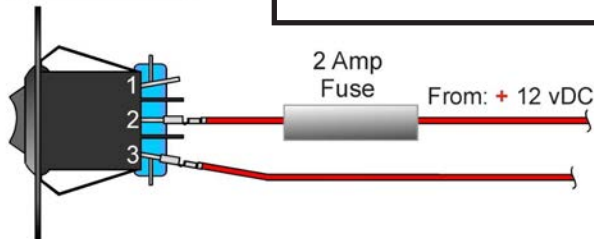
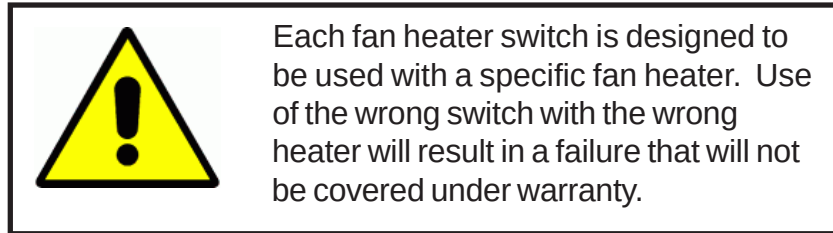


Electrical System

Fan switches for computer style (box) fans

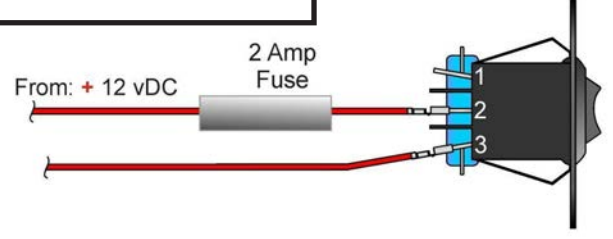
12 volt DCFan Heater

2 Speed Switch (No Off) Wiring Schematic



12 volt DCFan Heater

2 Speed Switch Wiring Schematic



1. Select a location for the switch near the fan heater where it will be easily accessible and easy to wire.
2. Cut a 1-1/2" H x 3/4" W hole for the switch.
3. Attach all electrical connections with spade connectors.
 - Terminal #2 is from the DC power source.
 - Terminal #3 is to the fan heater itself (red wire).
4. Center plate in hole and secure with 4 screws.

Fan aquastat kit #W005-213-REAL prevents fan from operating until temperature exceeds 110 degree F. Wire inline with the fan's ground wire (black). Switch is normally open and closes at 110 degree F. Several fans may be wired to the same switch up to 10 amps @ 12 volt DC. Mount as shown. Use provided thermal paste to ensure proper heat sensing.



3 Speed switch for blower (defrost) fans



Aquastat

To prevent cold air discharge from the fan heaters during initial startup, an inline temperature sensing aquastat is available to install in the return line. This aquastat keeps the fans off until a water temperature of 120° F has been reached.

Domestic Hot Water Aquastat

This aquastat is plumbed into the domestic hot water output with a tee at the water heater or in the case of an Everhot into the dedicated 3/4" FNPT coolant port. The domestic hot water aquastat acts like another thermostat. It senses the water temperature at the domestic water heater and tells the boiler when to turn on and off to keep the domestic water hot.

Circuit Breakers and Fuses

The main breaker is to protect the main power line, not the rest of the system. Each fan should be protected by an appropriately sized fuse or breaker.

Fan Relay

A fan relay allows a low power signaling device, in this case a thermostat, to control higher current or higher voltage devices. Each of these relays can operate one or more fans. This general purpose relay has contacts rated to 30 Amps at 12 VDC. A socket with spade connectors is available for this relay.

Master Switch

A "Master Switch with Indicator Light" allows you to quickly and clearly determine if your heating system is enabled.

This satisfies the ABYC requirement for a positive shut off with a visible system status indicator.



Optional switch plates labeled SYSTEM HEAT/ OFF/ ENGINE HEAT and SYSTEM HEAT/ OFF / ENGINE HEAT / ENGINE PREHEAT including an indicator light are available. Contact SMS for wiring and plumbing diagrams.

REAL™ Fans and Switches

REAL Fan Heaters are available in single or dual fan models. Multiple speeds can be obtained by using a fan switch. By running the already quiet computer fans on low, it makes them almost inaudible.

A single switch can be used on a dual fan model or you can control both fans independently with two switches. These switches are available with or without a positive off position. Use the "No Off" switch in areas controlled by a room thermostat. The thermostats will turn the fans on and off. Fan switches with the true "Off" position are used to control an area completely manually. These would be areas like a head, or small stateroom where there is no room thermostat actively monitoring the space.



It is important that the fan switches match the intended unit's voltage and amperage requirements. Ask SMS for help in sizing the proper switch.

REAL defrosters come with a squirrel cage blower that is internally wound for 3 speeds. These units come with a 3 speed fan switch.



Place the provided "IMPORTANT" warning sticker near the heating system master switch.

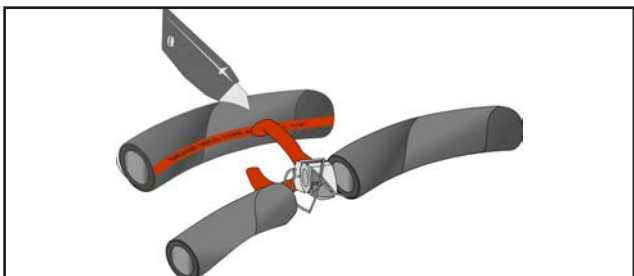


Water System

Hose

3/4" rubber hose is the most common method used for the installation of hydronic heaters. It is flexible, easy to install and the proper hose can last the life of the heater. Proper hose choice in this matter is key. Standard water hose and common automobile radiator hose do not work for the long runs required. If the hose is not specifically designed for heater installation, it will tend to kink or fail causing damage to your heating system and possibly to your boat.

All hose should be reinforced rubber hose. This type of hose will not soften with the operating temperatures of the heater. The reinforcement may be a fabric layer or wire. The case of the reinforced hose can be cut with a knife but wire reinforcement will require wire cutters.



All hose connections such as hose barbs or unions should be made of brass, copper or stainless steel. All hose clamps used to join the hose should be stainless steel. Further, SMS advises the use of double-banded clamps. Double-banded clamps protect the hose from being cut by the holes in the clamp itself.

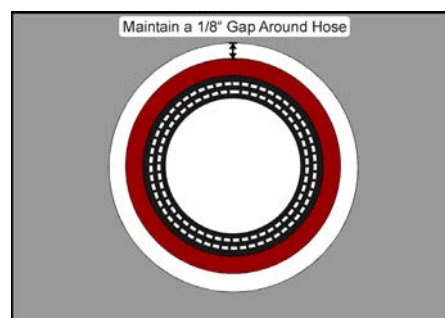
SMS recommends the use of a hose adhesive on the barb connections. *Especially important with beaded (rather than barbed) fittings, e.g., copper 90-degree elbows and the inlet/outlet fittings of the heater itself.



The hose should fit snugly and completely over the hose barb. The clamp should be tightened over the barb.



If you have to cut a hole to route a hose, the hole should be at least 1/4" of an inch larger in diameter than the hose. This will allow a 1/8" gap around the hose which will help avoid chaffing. This gap should exist even if a shield is installed for the hose to pass through.



The system hose should be supported by corrosion-resistant stainless steel or plastic clamp to a bulkhead or another stable surface. Nylon hose mounts are the most common and cost effective way to accomplish this.

If the system installation calls for a tight turn in the hose, a copper 90-degree elbow should be used instead of attempting to force the hose to make the tight turn. By forcing the hose, a kink may form and pinch off the system.



If the hose makes a high point and may trap air, an in-line bleeder or 90-degree bleeder should be installed.

A fill/drain valve with common garden hose thread should be installed at the lowest point in the system. This valve is to assist in the flushing, draining and filling the system for initial start up and periodic maintenance. Sure Marine Service offers this valve and any hardware necessary to install it into the system.

PEX

PEX (Cross-linked Polyethylene) has been used in potable water and for hydronic heating systems in Europe for over 30 years. PEX has replaced polybutylene (PB) as the most widely-used flexible plumbing pipe. We use PEX that is specifically made for radiant heating.

It is unaffected by electrolysis and compared to copper, installation should need fewer fittings without a need for torches, solder, flux, and so on.



To make a connection you slip a metal band over the tube, insert a fitting into the tube, then crimp in place using a crimping tool. The connection is ready for use. You can inspect the connection using a simple "Go/No-Go" gauge.

Copper Tube

Copper tube is a standard for household heating but space restrictions limit its effectiveness in the marine environment.

Copper tube is used in installations that have long, straight runs. Because of the nature of boats in general, there are few long, straight runs of anything. Bends require soldering which, because of proximity to combustibles, cannot be done safely inside the boat. This requires the construction of complex pieces of tube outside the boat prior to installation. This is not the best method for a tight installation and many installers opt for hose or PEX tubing.

Copper tube can, however, be used in cases in which a small amount of heat should come from the system without the use of fan heaters or tube fin. Exposed copper tube is rated to 250 BTU per foot. An exposed length in a locker or lazarette can provide heat to dry clothes or gear. These exposed pieces should be protected and away from high traffic areas. Water flowing through these tubes is up to 180 degrees Fahrenheit and can burn exposed skin.

Boiler

The boiler has two ports for plumbing connections, Supply (discharge) and Return (from suction pump). The lower fitting away from the burner head is the Return.

If the water outlet exits the heater and turns down a tee should be installed with a bleeder on top of the connection.

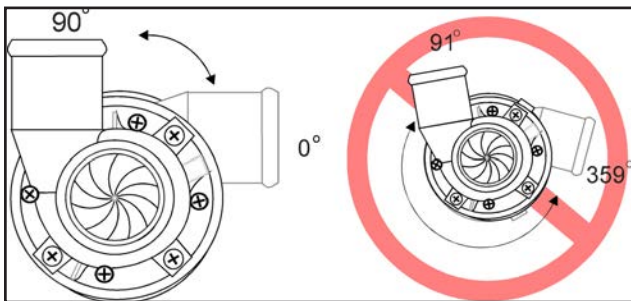


Pump

Water flow through the system is very important and trapped air is the most common impediment to the proper flow. The system pump can be a source of air in a system if it is not properly installed. It is a centrifugal pump and it must have head pressure on the suction side to operate properly and it must be installed in a particular manner to move water without generating air bubbles. You are using water to distribute heat, bubbles won't work.

Maximum recommended single loop hose run is 100'.

Most Centrifugal pumps are designed with center suction and side discharge. The side discharge must face between 0 and 90 degrees. Installation outside of this parameter will result in cavitation of the pump and put bubbles in the system.

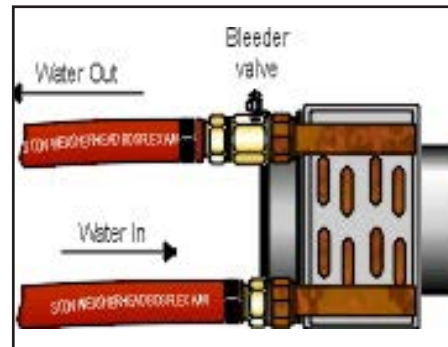


The pump must be installed horizontally. If the pump is installed at an angle or vertically, it can trap air in the head that will adversely affect water flow

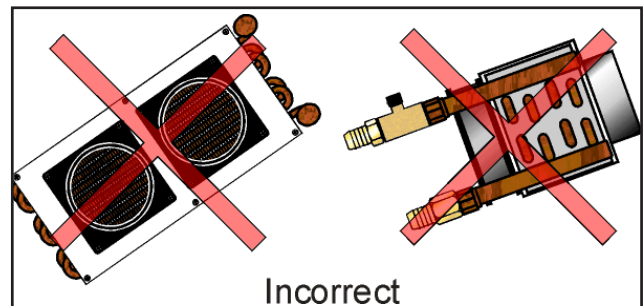


REAL™ Fan Heaters

Before installing the fan heater, make sure that the space selected will meet the manufacturer's specifications. Every fan heater should have a bleeder on the outlet side. If you can not locate an air bleeder, contact the manufacturer or SMS for options.

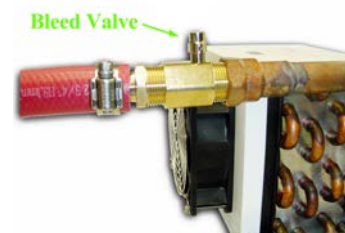


Water enters at the bottom connection and flows out from the top connection (unless otherwise specified by the manufacturer).



No matter the make or model, a fan heater must be installed level. Some brands allow for several installation configurations, but all are on a flat surface. An angled installation will trap air such that it will not reach the bleeder.

Once the system is installed and filled, the fan heater must be bled. This can be accomplished by opening the bleed screw on the discharge side of the fan heater and leaving it open until a steady stream of water flows out. If the water is spitting out of the fitting, air is still present. See the "bleeding the system" section.

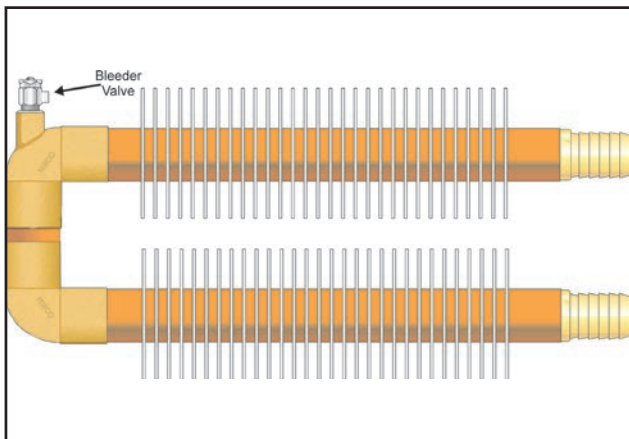


Tube fin

Tube fin is a copper tube run through aluminum sheets designed to convect heat into the boat. This is the same system as a household baseboard heating system. Tube fin is advantageous because it does not draw any power from the batteries, however it requires considerably more room than a fan heater.

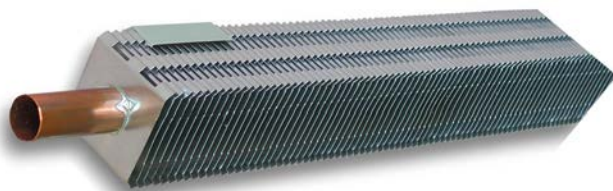
Most tube fin does not exceed 750 BTU per foot (3 x 3 inch) and commonly type is only 500 BTU per foot (2 x 2 inch).

While tube fin is convenient for small lockers and heads, several feet are required to heat larger spaces. Because of the angles of a boat, a long single run of tube fin is rarely feasible. Of course you can combine them with fan heaters in tight areas.



Two (but no more) lengths can be stacked on top of each other. If stacked tube fin is installed, a bleeder should be installed on the top tube.

Tube fin should be installed low in the cabin to allow for the heat to rise. A guard for tube fin should be fabricated to fit your set up. Standard household baseboard guards are available for single-pass tube fin, but for stacked systems it may be easiest to have a guard fabricated to fit.



Domestic Water Heaters

The traditional domestic water heater is a tank of potable water heated by a loop from the heating system or by an electrical coil. Consult your water heater installation manual for mounting, wiring and plumbing specifications.

Most domestic water heaters have a 5/8" coil passing through the tank. A 3/4" bypass valve can be installed to allow water to pass through the tank while maintaining full heating system flow.

The bypass valve for the domestic water tank must be capable of throttling. SMS recommends using a gate valve for this purpose.

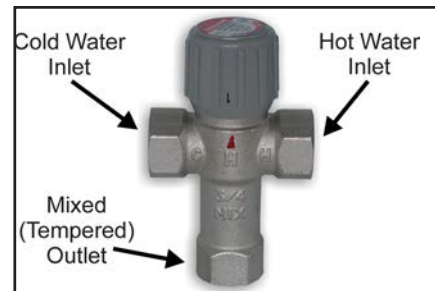


Everhot Water Heater

The Everhot is an on-demand domestic water heater. Instead of a tank that stores water, the Everhot heats water by passing it through a coil of finned copper tubing surrounded by 180° F coolant from the boiler. Domestic water is heated up to 170 degrees F in one pass. The temperature rise is enough to require a mixing valve to avoid producing scalding hot water.



A mixing valve must be installed in the Everhot domestic water system to avoid scalding.



The Everhot has 1 1/4" boiler fittings for the water jacket and 3/4" or 1/2" fittings for the domestic water. Thanks to the full size boiler fittings and the low resistance within the water jacket an Everhot can be installed in the main water line or on a 3/4" loop. If installing in a loop, be sure that loop will be open at all times.

Expansion Tank



The expansion tank must be located at the highest point of the heating system and connected as directly as possible to the inlet of the circulation pump.



This kind of system must allow for expansion and contraction of system components as they heat up and cool down. The expansion tank accommodates this by allowing the water level to rise and fall while still maintaining head pressure on the system pump. At the same time, the expansion tank acts as a “burp” tank, allowing for air in the system to escape.

If any component, such as a fan heater, sits higher than the expansion tank, it will (essentially) take over that function. Air will travel to the highest point in the system and stay there. For example, if a defrost fan heater were installed above the expansion tank, air would trap in the fan heater and it would most likely not perform to specifications. Further, the air would bind the water flowing through the fan heater and plug the loop in which it is installed, leading to failure of other components in the system.

The expansion tank may be installed anywhere in the system provided it is the highest point and there is no restriction between it and the system pump. It is possible and in most cases desirable to install the expansion tank on one of the loops from the manifold. The loop with the expansion tank should be marked to warn against shutting it off completely. One side can be closed to stop circulation but still allow for expansion.

Engine Heat Exchanger

Follow your engine manufacturer's guidelines in regard to connecting an engine heat exchanger to your engine's fresh water cooling system. Do not connect this heat exchanger to raw water cooled engines. The engine's manufacturer should be able to tell you which specific ports on the engine to use for auxiliary heat.

The heat exchanger should be mounted close to the engine since the engine's fresh water cooling pump is not designed to push the coolant great distances.

The heat exchanger must be mounted below the engine's coolant reservoir.

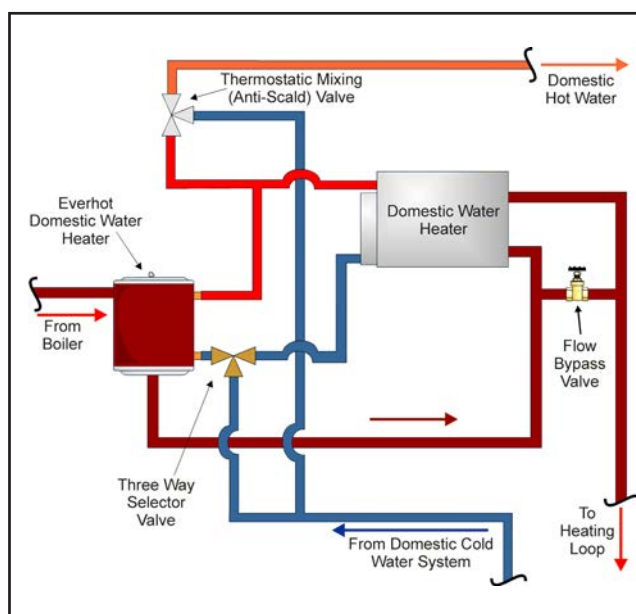
The engine coolant hoses on the heat exchanger must exit on the bottom of the heat exchanger and then loop back up to the engine to avoid thermal cycling, otherwise you may constantly heat your engine block and this may damage your engine. Preheating your engine for easy start up is okay if you use an engine preheat pump installed between the engine and the engine heat exchanger. This pump should be controlled with an on/off switch and only run long enough to warm the engine block prior to starting. Mounting the heat exchanger with the engine fittings pointing down will also ensure that the air bleed screw is on the top of the heat exchanger.



Everhot/Domestic Hot Water Tank Combination

Both types of domestic water heat have their advantages. While an Everhot can provide an endless supply of hot water, it requires the operation of your boiler. A standard domestic water heater provides hot water independent of the heating system but it has a limited capacity of hot water and requires generator or shore power.

An Everhot and a standard domestic water heater can be installed parallel to each other in the domestic water system. Although this system requires more space, the advantages of either system can be used with the turn of a valve.



A three-way valve is required to use as a selector between the systems. This valve should be installed in a readily accessible area and clearly labeled so you can tell which system is selected.



Water Filling Guidelines

The component listing below can help you figure the volume of water your heating system holds.

During normal operation the system should have a 60/40 water/antifreeze mix.

Multiply the capacity of your system by 0.4 to determine how much antifreeze you will need.

For example if you calculated that your system holds 5 gallons then you will need 2 gallons of antifreeze ($5 \times 0.4 = 2$).

In this case have someone get 2 gallons of low silicate antifreeze but don't add it yet! Test for leaks FIRST.

COMPONENT	CAPACITY
Webasto Boiler	0.4 Qt.
Everhot	5.7 gallons
REAL Fan Heaters	0.2 quart ea.
3/4" Heater Hose	2.97 gal per 100 ft.
3/4" Copper Tube	2.29 gal per 100 ft.
1 1/4" Flex hose	6.4 gal per 100 ft.
5 gal Expansion tank	5.2 gallons (full)

Fill - Check - Flush

The system should be flushed to expel any hose shavings, sawdust and so on that may have been introduced during installation.



First remove the expansion tank's cap to allow air to escape. Connect city water to the low point fill/drain valve and fill the system until water is to the 1/4-full mark in the expansion tank.

Allow city water pressure to push water through the system.

Do not energize your system pump.



Any foreign object in the system can foul or damage the impeller.

While the system is filling you should be checking for leaks. You want to find any leaks or kinks at this point instead of when the system has been filled with antifreeze. Once the inspection is completed, drain the water from the system.

The water pump should never be run when the system is dry.



If you noticed any leaks or kinks you will want to do this again after you make your correction. Remember, at this point to use water only because you will be draining the system again.

Bleeding the Water System

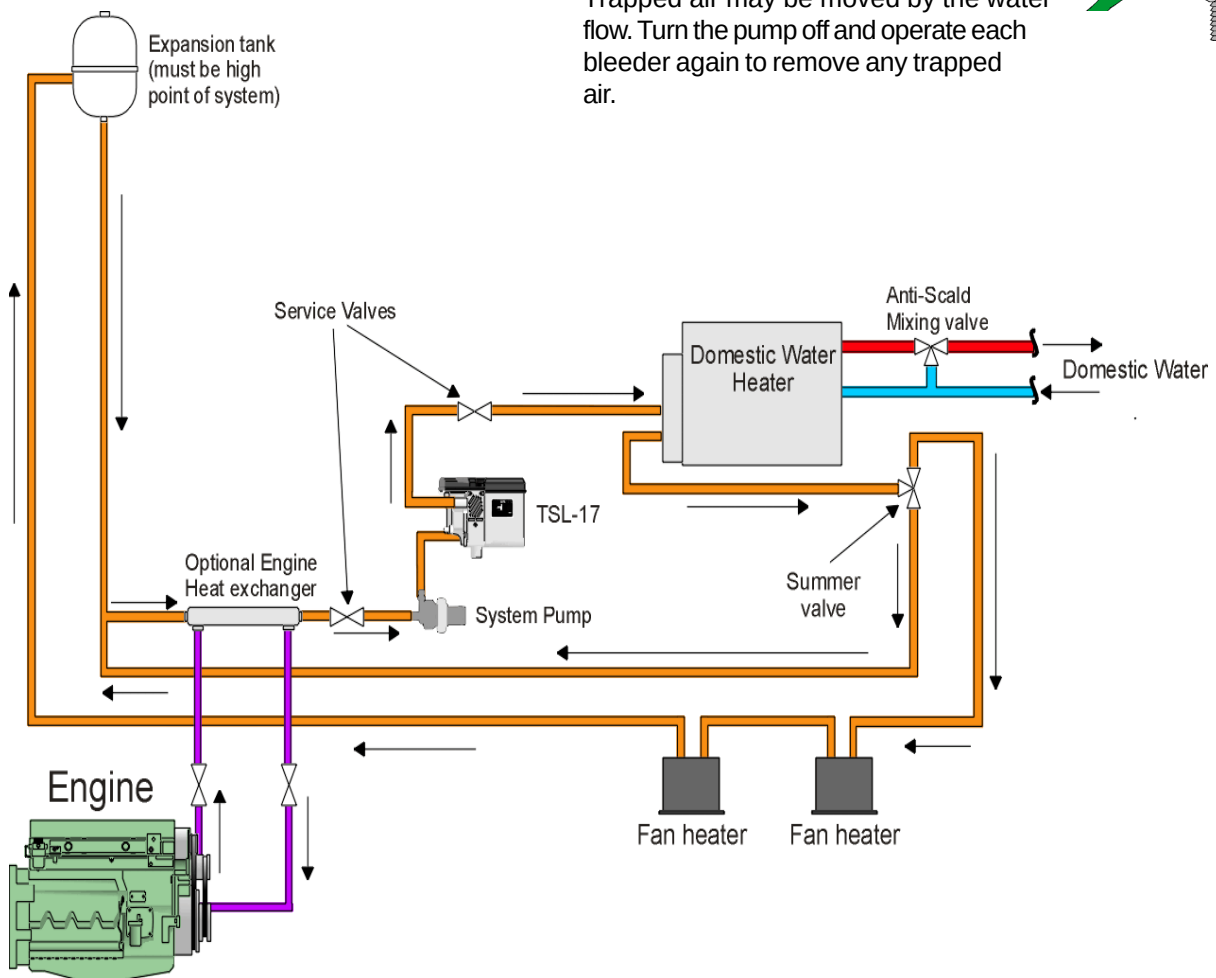
After flushing, use water to refill the system to the $\frac{1}{4}$ level in the expansion tank.

The water system should be bled before any operational tests, initial start up, for periodic maintenance and if there is suspicion of air trapped in the system.

Each fan heater should have a bleeder, as well as all high-point loops.

Make sure the system is cool and remove the expansion tank cap. Open each bleeder in turn and allow water to flow out. If the water sputters, there is air present. The bleeder should be left open until a constant stream of water exits.

To make sure the system has been bled properly, energize the pump for a short time. Trapped air may be moved by the water flow. Turn the pump off and operate each bleeder again to remove any trapped air.



Commissioning

Water System Check

This test should be done after the water system has been flushed, all other systems have been properly installed, and the entire system is ready for start up.

Fill the system with water only and bleed, then run an operational test.

Test with water only in order to minimize damage or pollution if the system leaks when up to temperature, or if there is an unforeseen water system failure.

Water alone should never be used as the operational coolant.



Using only water will lead to corrosion in the system and will shorten the operational life of the components in the system.

With water only in the system, send power to the system and turn all thermostats up.

As the boiler begins to heat the system, inspect the entire installation. The boiler must have no oil, exhaust, or water leaks. Check all hose for kinks and all fittings for leaks.

Continue your inspection until the system is running at full operational temperature.

With an aquastat in the system, the fan heaters will not turn on until the coolant has warmed sufficiently.

When the system is fully operational there should be a 10°F difference between the coolant return and supply at the boiler. This difference is called the ΔT (Delta T).

If the ΔT exceeds 17°F, coolant flow is too low and the system will not function properly.

When you have completed your test turn all thermostats down and shut the system down to allow the water system to cool.

After the system has cooled down, check the water level in the expansion tank. If the system has lost a

substantial amount of water, there may be a leak in the system. After the leak has been located and fixed, another inspection should be conducted.

Never check the system water level when the system is hot.

Checking a hot system may result in scalding from escaping water or steam.



Once the system has been cleared for any leaks, antifreeze should be added.

The operating water system should be a 60% water 40% antifreeze mix. This is to avoid freezing of the system and ensure lubrication of the pump. The antifreeze should be a low-silicate antifreeze. Standard auto antifreeze contains high silicate counts that may leave deposits in the system and cause flow problems.

The operating water system should never exceed 50/50 water to antifreeze mix.

Using the total volume of the system figured before the test, drain 40% of the water in the system. For example, if you calculated that your system holds 21 and 1/4 gallons of water then you would want to drain out 8 and 1/2 gallons of water ($21.25 \times 0.4 = 8.5$).

Close the drain and add antifreeze until you reach the $\frac{1}{2}$ full level in the expansion tank. Leaving the expansion tank cap off, energize the system pump, but not the heater. Bleed the system of any air.

Once the system has been bled, the level in the expansion tank may have gone down. Add more antifreeze and bleed the system until the expansion tank is half full when the system is cool.

The coolant should be maintained at a PH factor of 8 to 9. If the coolant becomes acidic (lower than 7 on the PH scale) corrosion of the metal components within the system can be expected. Periodically the PH level of the coolant should be checked and adjusted with a coolant additive if below 8 on the PH scale.



Sealants and their Uses

Your system may have come with a number of sealants. Each sealant has a specific use.

Water System Sealants

A leak in your coolant system is an annoyance and can require a lot of time to track down and fix. Using the proper sealants can greatly reduce the possibility of leaks.

Permatex Form-A-Gasket #3



SMS recommends the use of a hose adhesive on the coolant* hose connections. This is especially important on the beaded style fittings that come out of the heater as they do not “grab” the hose as well as the barbed style fittings.

*Do not use this product on fuel hoses.

How to use:

1. Apply thin, even coats to the inside of the hose and the outside of the hose fitting.
2. Allow to air-dry several minutes before assembling.

A hose clamp will still be required.

FasSeal-ATS



FasSeal-ATS is an anaerobic pipe thread sealant with PTFE (teflon). Use on all metal to metal pipe thread (NPT) joints. Forms a tough, insoluble, vibration-proof seal.

How to use:

1. Clean fittings to remove excess oil, dirt, and grease.
2. On small diameter pipes apply to male threads only; for larger diameters, apply to both male and female threads.
3. Assemble and tighten joint.

Fuel System Sealants

FasSeal-ATS (see above) may also be used on fuel fitting pipe thread (NPT) joints.

Exhaust System Sealants



A leak in your exhaust system can create a life threatening condition. Special care must be taken to avoid any exhaust leaks. All fittings in the exhaust system must be properly sealed and clamped to prevent any leakage of exhaust gasses into the boat's living spaces.

Permatex Ultra Copper



Ultra Copper* is a high temperature RTV Silicone sealant resistant to temperatures up to 700 degrees F. Use this product to seal the exhaust tubing to the heater and exhaust thru-hull. Also use this product to seal the exhaust thru-hull gaskets to prevent water intrusion at the thru-hull.

* Despite the name, Ultra copper contains no copper and is safe to use on aluminum.

How to use - Exhaust Tubing:

1. Clean and dry all surfaces to be sealed.
2. Apply cement only to the male side of the fitting.*
3. Assemble the joint and clamp. Allow the joint to dry gradually. Do not move or twist after joining as this will break the seal.
4. Clean any excess sealant with a non-flammable cleaning fluid.
5. Silicone dries in approximately 1 hour, and fully cures in 24 hours.



* Avoid getting excess sealant in the flex tube. If the sealant is applied to the female side, it may impede the flow of exhaust gasses and cause serious damage to the unit.

How to use - Exhaust Gaskets:

Drill hole and install thru-hull according to the instructions included with the thru-hull

1. Clean and dry all surfaces to be sealed.
2. Liberally apply a coat of high temperature gasket sealer to both sides of each gasket.
3. Install thru-hull making sure to maintain an 1/8" gap around the thru-hull and the hole you drilled into the boat.
4. Clean any excess sealant with a non-flammable cleaning fluid.
5. Silicone dries in approximately 1 hour, and fully cures in 24 hours.



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